

97.2% Prediction Accuracy (R ²)	±0.41 days Avg Prediction Error	15 Reassignment Opportunities	10,194 Orders Analyzed
---	---	---	----------------------------------

The Situation

Nassau Candy currently assigns its 15 products to 5 factories using static, legacy rules with no mechanism to simulate the impact of a reassignment before it happens. This creates three recurring problems: suboptimal shipping distances, elevated lead times for certain regions, and avoidable margin erosion from logistics inefficiency.

What We Built

A machine learning system that predicts shipping lead time for any product–factory–destination combination and recommends optimal factory reassignments. Three models were tested; the best-performing model (Gradient Boosting) explains 97.2% of lead time variance and predicts within ±0.41 days of actual outcomes accurate enough to confidently distinguish between factory options.

- Live interactive dashboard for leadership and operations teams to explore scenarios in real time
- Automated recommendation engine ranking all 15 products by reassignment opportunity
- Risk flagging that protects profit margin by penalising long-distance reassignments

Key Findings

Finding	Business Implication
Lot's O' Nuts (Arizona) is the slowest origin for East Coast orders	3 Chocolate products show ~2.6–2.7 day savings if reassigned to Secret Factory (Illinois)
Sugar Shack (Minnesota) underperforms on southern routes	Laffy Taffy alone could save ~2.5 days by shifting to The Other Factory (Tennessee)
Wicked Choccy's (Georgia) is the network's most efficient factory	Best candidate to absorb additional Atlantic / South-East volume
No high-risk reassignments identified	All 15 recommended moves are financially safe, none significantly increase shipping distance

Recommended Actions

- Approve reassignment of Lot's O' Nuts Chocolate products (Fudge Mallows, Scrumdiddlyumptious, Nutty Crunch Surprise) to Secret Factory for East Coast orders highest aggregate impact.
- Evaluate Sugar Shack production capacity before shifting Laffy Taffy to The Other Factory.
- Use the What-If Analysis tool to validate each reassignment against specific customer accounts before execution.
- Re-run the Recommendation Dashboard quarterly as order volumes shift seasonally.

Access the System

Live dashboard:

<https://candy-factory.streamlit.app>

Full technical methodology, model evaluation, and detailed findings are documented in the accompanying Research Paper.